MTH636 – Game Theory

Lecture 24: Behavior Strategies, Mixed Strategies & Equivalence

1 Complexity of Behavior vs. Mixed Strategies

Recall that Player i has k information sets, and the ℓ-th information set has m_ℓ many actions.

- For a **behavior strategy** b_i, we have to specify

$$\sum_{\ell=1}^{k}(m_\ell - 1) \quad \text{numbers.}$$

- For a **mixed strategy** σ_i, we have to specify

$$\left(\prod_{\ell=1}^{k} m_\ell - 1\right) \quad \text{numbers.}$$

Note

Behavior strategies are *exponentially* more compact than mixed strategies, since $\sum_\ell(m_\ell - 1) \ll \prod_\ell m_\ell - 1$ in general for large k.

2 Equivalence Definition

Definition: Equivalence of Mixed and Behavior Strategies

A mixed strategy σ_i and a behavior strategy b_i of Player i in an extensive form game Γ are said to be **equivalent** if, for every mixed/behavior strategy vector σ_{-i} of the other players and for all vertices n of the game tree,

$$\rho\Big(n;\,(\sigma_i, \sigma_{-i})\Big) = \rho\Big(n;\,(b_i, \sigma_{-i})\Big),$$

where $\rho(n;\cdot)$ denotes the probability of reaching vertex n under the given strategy profile.

2.1 Illustrative Game Tree

Consider the following three-player extensive form game tree (Player I at root n_0, Player II at n_1, and Player I again at the information set $\{n_3, n_4\}$):

Tree structure:

- n_0 (Player I): actions A (to n_1) and B (to n_2).

- n_1 (Player II): actions L (to n_3) and R (to n_4). Nodes n_3 and n_4 belong to the same information set of Player I (dashed box).
- At the information set $\{n_3, n_4\}$ (Player I): actions ℓ (to n_5 or n_7) and r (to n_6 or n_8).
- Terminal nodes: n_2, n_5, n_6, n_7, n_8.

3 Remarks on Equivalence

Remark R-1

Suppose b_i (a behavior strategy) and σ_i (a mixed strategy) are equivalent. Then, for all mixed/behavior strategy vectors σ_{-i} of the other players $\neq i$,

$$U_j(\sigma_i, \sigma_{-i}) = U_j(b_i, \sigma_{-i}).$$

More explicitly, this gives:

$$U_j\Big(\sigma_i, \sigma_j, \sigma_{-\{i,j\}}\Big) = U_j\Big(b_i, \sigma_j, \sigma_{-\{i,j\}}\Big).$$

Remark R-2

Let $\sigma = (\sigma_i)_{i\in N}$ be a mixed strategy profile and $b = (b_i)_{i\in N}$ be a behavior strategy profile. Suppose b_i is equivalent to σ_i for all i. Then we have

$$U_i(\sigma) = U_i(b) \quad \forall\, i.$$

4 Worked Example: Verifying Equivalence

4.1 Setup

Consider the game tree with:

- n_0 (Player I, root): actions A (prob. $\frac{1}{2}$) and B (prob. $\frac{1}{2}$) under mixed strategy σ_1.
- n_1 (Player II): actions L and R.
- Information set $\{n_3, n_4\}$ (Player I): actions ℓ and r.
- Behavior strategy of Player II: $\sigma_2(L) = 1$.

Under the mixed strategy σ_1 (Player I plays A with prob. $\frac{1}{2}$ and B with prob. $\frac{1}{2}$; within B, plays ℓ with prob. $\frac{1}{3}$ and r with prob. $\frac{2}{3}$):

$$\sigma_1(A\ell) = \tfrac{1}{2} \times \tfrac{1}{3}, \qquad \sigma_1(Ar) = \tfrac{1}{2} \times \tfrac{2}{3},$$
$$\sigma_1(B\ell) = \tfrac{1}{2} \times \tfrac{1}{3}, \qquad \sigma_1(Br) = \tfrac{1}{2} \times \tfrac{2}{3}.$$

The reaching probabilities must satisfy:

$$\rho\Big(n_2;\, (\sigma_1, \sigma_2)\Big) = \rho\Big(n_2;\, (b_1, \sigma_2)\Big),$$

$$\rho\Big(n_i;\, (\sigma_1, \sigma_2)\Big) = \rho\Big(n_i;\, (b_1, \sigma_2)\Big) \quad \forall\, n_i \in L,$$

where L denotes the set of all leaves.

4.2 Constructing an Equivalent Mixed Strategy from a Behavior Strategy

Given a behavior strategy b_i, we can construct an equivalent mixed strategy σ_i via:

$$\sigma_i(s_i) = \prod_{h_i \in H_i} b_i\Big(s_i(h_i),\, h_i\Big),$$

i.e., the probability of a pure strategy s_i under σ_i is the product of the behavior strategy probabilities at each information set.

Example. Using the tree above with actions A, B, ℓ, r and probability labels:

- Probabilities at n_5, n_6: $\ell \mapsto \frac{1}{3}$, $r \mapsto \frac{2}{3}$.
- Probabilities at n_7, n_8: $\ell \mapsto \frac{1}{2}$, $r \mapsto \frac{2}{3}$.

Computing reaching probabilities:

$$\sigma_1(A\ell) = 0, \quad \sigma_1(Ar) = 0, \quad \sigma_1(B\ell) = \tfrac{1}{2} \cdot \alpha, \quad \sigma_1(Br) = \tfrac{1}{2}(1-\alpha),$$

for some free parameter $\alpha \in [0, 1]$.

Note

This equivalence is **not unique** in the sense that given a behavior strategy, there can be **multiple** mixed strategies which are equivalent to it. The parameter α above is free, illustrating this non-uniqueness.

5 Theorem: Equivalence Always Exists (Behavior ⇒ Mixed)

Theorem: Existence of Equivalent Mixed Strategy

Given a behavior strategy b_i, there always exists an equivalent mixed strategy σ_i.

Proof. Given b_i, construct σ_i by setting

$$\sigma_i(s_i) = \prod_{h_i \in H_i} b_i\Big(s_i(h_i),\, h_i\Big).$$

One then verifies that this σ_i produces the same reaching probability at every vertex of the game tree as b_i does. □

6 Mixed ⇒ Behavior: The Non-Trivial Direction

The non-trivial question is: given a mixed strategy σ_i, does an equivalent behavior strategy exist?

6.1 Example

Consider a game tree where Player I has two information sets $h_1 = \{n_0\}$ (root) and $h_2 = \{n_1, n_2, n_3\}$ (a three-node information set). Player I has pure strategies $A\ell$, Ar, $B\ell$, Br, and the mixed strategy satisfies:

$$\sigma_1(A\ell) = \tfrac{1}{2}, \qquad \sigma_1(B\ell) = \tfrac{1}{2},$$
$$\sigma_1(Ar) = 0, \qquad \sigma_1(Br) = 0.$$

Let $b_1(A;\, n_0) = u$ (probability of action A at root) and $b_1(\ell;\, h_2) = y$ (probability of action ℓ at h_2). Then:

$$\rho(n_3;\, \sigma_1) = \tfrac{1}{2} = uy, \qquad 0 = (1-u)y,$$
$$\rho(n_4;\, \sigma_1) = 0 = u(1-y), \qquad \tfrac{1}{2} = (1-u)(1-y).$$

From the system $uy = \frac{1}{2}$ and $(1-u)y = 0$: we get $y = 0$ (from the second equation if $u \neq 1$) but $uy = \frac{1}{2}$ requires $y > 0$. This contradiction shows the system has **no solution**, demonstrating that:

Key Insight: No Equivalent Behavior Strategy Without Perfect Recall

If we do **not** have *perfect recall*, then there exists a mixed strategy which does **not** have an equivalent behavior strategy.